



CHAPTER

SonixED Stack Modules

The electro dialysis stacks at the heart of the unit

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10.1 Overview and function

The SonixED™ stack is the unit's core: an assembly of alternating cation- and anion-exchange membranes, titanium electrodes and spacers, with integrated ultrasound transducers for chemical-free antifouling. Under an applied DC voltage, ions migrate across the membranes from the diluate stream into the concentrate stream, desalinating the diluate. Each skid carries 1–3 stacks (standard configurations 1SS/2SS/3SS).



Figure 10.1. Suggested figure: labeled photograph of a single SonixED™ stack module, showing the electrode housings, membrane stack, ultrasound transducers and 1" BSP connections.



Figure 10.2. Suggested figure: a cutaway/exploded diagram of one cell pair, showing the diluate/concentrate/electrolyte flow paths through alternating membranes — useful for explaining the process to a new operator.

10.2 Key specifications

- Module dimensions (l×b×h): 300 × 400 × 1060 mm; weight ~150 kg; connection 1" BSP.
- Flowrate per stack: 0.5–10 m³/h (skid datasheet); 24 m³/day (stack datasheet).
- Salt rejection: up to 90%; water recovery: up to 98%.
- Operating pressure: 0.5–2.5 bar (typical 0.5–1 bar); maximum 3 bar.
- Temperature: 5–40 °C (stack datasheet) / 5–55 °C (skid datasheet) — TODO reconcile which applies to the finished product.
- DC voltage across stack: 60–600 V; DC current: 3–20 A.
- Cell pairs per stack: 50–250; effective membrane area: 10–50 m² ("long" variant) or 20–100 m² ("square" variant).
- Membrane: Fujifilm Type 10, polyolefin.
- Feed water requirements: salinity ≤90,000 ppm; pH 5–8.5 typical (tolerable 2–13); turbidity <1 NTU; suspended solids <20 µm; SDI(5 min) 10; iron <0.5 mg/l; free chlorine 0.5 ppm continuous / 30 ppm intermittent tolerance.

10.3 Operation

Stacks operate continuously under PLC control once commissioned (chapter 5); electrodialysis *reversal* periodically swaps stream polarity as part of the ultrasound antifouling strategy — TODO confirm the reversal interval/trigger. Process values (pressure, flow, conductivity per stream) are monitored via the sensors described in chapter 13.

10.4 Maintenance

The membrane is the stack's principal wear item (weight ~150 kg, dimensions and connection as in *Key specifications* above). TODO: part number, reorder code and recommended replacement interval were not found; no CIP or cleaning procedure specific to the SonixED™ stack was found among the documents reviewed.

10.5 Troubleshooting

If salt rejection or recovery drops noticeably below the rated ranges above, check feedwater against the operating envelope in *Key specifications* first — turbidity, iron and free chlorine excursions are the most likely root causes of fouling-related performance loss. TODO: no documented fault table linking specific water-quality deviations to specific symptoms (e.g. “high iron causes X”) was found beyond the acceptable operating envelope itself.